DemandIQ — Entity Relationship Diagram

SQL Server · 27 tables · 3NF · PK primary key · FK foreign key · UQ unique · crow's foot = many, bar = one, circle = optional

Access & permissions

screens

PK	**screen_id**	VARCHAR(6)
	name	NVARCHAR(60)
UQ	path	VARCHAR(40)
UQ	position	SMALLINT

permission_actions

PK	**action**	NVARCHAR(40)
FK	screen_id	VARCHAR(6)
	allows_view	BIT
	allows_review	BIT
UQ	position	SMALLINT

role_permissions

PK,FK	**action**	NVARCHAR(40)
PK,FK	**role_id**	SMALLINT
	level	VARCHAR(6)

role_screens

PK,FK	**role_id**	SMALLINT
PK,FK	**screen_id**	VARCHAR(6)

roles

PK	**role_id**	SMALLINT
UQ	name	NVARCHAR(40)

report_types

PK	**name**	NVARCHAR(60)
	description	NVARCHAR(200)
	icon	VARCHAR(20)
UQ	position	SMALLINT

role_report_types

PK,FK	**role_id**	SMALLINT
PK,FK	**report_type**	NVARCHAR(60)

Sales data & uploads

datasets

PK	**dataset_id**	VARCHAR(12)
	file_name	NVARCHAR(260)
	uploaded_at	DATETIME2(3)
FK	uploaded_by	VARCHAR(12)
	total_rows	INT
	valid_rows	INT
	combined_rows	INT
	missing_rows	INT
	duplicate_rows	INT
	invalid_date_rows	INT
	invalid_numeric_rows	INT
	error	NVARCHAR(1000) NULL
	coverage_first_date	DATE NULL
	coverage_last_date	DATE NULL
	coverage_products	INT NULL
	cleaning	VARCHAR(12)
	cleaned_at	DATETIME2(3) NULL
FK	*cleaned_by*	VARCHAR(12) NULL
	outliers_capped	INT NULL
	records_replaced	INT NULL
	records_skipped	INT NULL
	records_added	INT NULL

dataset_pending_records

PK	**record_id**	VARCHAR(40)
FK	dataset_id	VARCHAR(12)
	sale_date	DATE
FK	sku	VARCHAR(40)
FK	location_id	INT
	units	INT
	unit_price	DECIMAL(12,2)
	discount_rate	DECIMAL(6,4)
	net_revenue	DECIMAL(14,2) computed

sales_records

PK	**sale_id**	VARCHAR(40)
	sale_date	DATE
FK	sku	VARCHAR(40)
FK	location_id	INT
	units	INT
	unit_price	DECIMAL(12,2)
	discount_rate	DECIMAL(6,4)
	net_revenue	DECIMAL(14,2) computed
FK	*dataset_id*	VARCHAR(12) NULL

Catalog, locations & inventory

regions

PK	**region_id**	INT IDENTITY
UQ	name	NVARCHAR(60)

locations

PK	**location_id**	INT IDENTITY
UQ	name	NVARCHAR(80)
FK	region_id	INT

categories

PK	**category_id**	VARCHAR(10)
UQ	name	NVARCHAR(80)
	description	NVARCHAR(200)
	status	VARCHAR(10)

products

PK	**sku**	VARCHAR(40)
	name	NVARCHAR(200)
FK	category_id	VARCHAR(10)
	channel	NVARCHAR(40)
	unit_price	DECIMAL(12,2)
	unit_cost	DECIMAL(12,2) NULL
UQ	position	INT
FK	*added_from*	VARCHAR(12) NULL

inventory

PK,FK	**sku**	VARCHAR(40)
FK	location_id	INT
	stock	INT
	reorder_level	INT
	safety_stock	INT
	demand_30d	INT
	updated_at	DATETIME2(3)

inventory_snapshot_items

PK,FK	**month**	VARCHAR(7)
PK,FK	**sku**	VARCHAR(40)
	stock	INT
	status	VARCHAR(10)

inventory_snapshots

PK	**month**	VARCHAR(7)
	as_of	DATE

reorders

PK	**reorder_id**	VARCHAR(12)
FK	sku	VARCHAR(40)
	quantity	INT
	supplier	NVARCHAR(120)
	status	VARCHAR(10)
	ordered_at	DATETIME2(3)
	delivered_at	DATETIME2(3) NULL
FK	ordered_by	VARCHAR(12)

Accounts & audit

users

PK	**user_id**	VARCHAR(12)
	name	NVARCHAR(120)
UQ	email	NVARCHAR(254)
	password_hash	VARCHAR(100)
FK	role_id	SMALLINT
	department	NVARCHAR(60)
	status	VARCHAR(20)
	created_at	DATETIME2(3)

sessions

PK	**session_id**	VARCHAR(12)
FK	user_id	VARCHAR(12)
FK	role_id	SMALLINT
	login_at	DATETIME2(3)
	logout_at	DATETIME2(3) NULL
	expires_at	DATETIME2(3)
	remember	BIT
	token_hash	VARCHAR(64) NULL

activity_log

PK	**event_id**	VARCHAR(12)
	created_at	DATETIME2(3)
FK	*session_id*	VARCHAR(12) NULL
FK	*user_id*	VARCHAR(12) NULL
FK	*role_id*	SMALLINT NULL
	attempted_login	NVARCHAR(254) NULL
	action	NVARCHAR(80)
FK	module	NVARCHAR(40)
	details	NVARCHAR(400)
	status	VARCHAR(10)

modules

PK	**name**	NVARCHAR(40)
UQ	position	SMALLINT

error_log

PK	**error_id**	VARCHAR(12)
	created_at	DATETIME2(3)
FK	module	NVARCHAR(40)
	error_type	NVARCHAR(60)
	severity	VARCHAR(10)
	message	NVARCHAR(1000)
	status	VARCHAR(15)
	notes	NVARCHAR(1000) NULL

Forecasting

forecast_runs

PK	**forecast_id**	VARCHAR(12)
FK	sku	VARCHAR(40)
	model	VARCHAR(20)
	horizon	INT
	start_date	DATE
	end_date	DATE
	window_size	INT NULL
	slope	DECIMAL(14,4) NULL
	arima_p	SMALLINT NULL
	arima_d	SMALLINT NULL
	arima_q	SMALLINT NULL
	mae	FLOAT NULL
	rmse	FLOAT NULL
	mape	FLOAT NULL
	holdout_days	INT
	generated_at	DATETIME2(3)
FK	*generated_by*	VARCHAR(12) NULL

forecast_sources

PK,FK	**forecast_id**	VARCHAR(12)
PK	**kind**	VARCHAR(7)
PK	**position**	SMALLINT
FK	*dataset_id*	VARCHAR(12) NULL
	records	INT
	first_date	DATE
	last_date	DATE

forecast_points

PK,FK	**forecast_id**	VARCHAR(12)
PK	**period**	INT
	predicted	INT

Reports

reports

PK	**report_id**	VARCHAR(12)
FK	report_type	NVARCHAR(60)
	format	VARCHAR(4)
	generated_at	DATETIME2(3)
	completed_at	DATETIME2(3) NULL
FK	user_id	VARCHAR(12)
FK	role_id	SMALLINT
	status	VARCHAR(10)
	filters	NVARCHAR(MAX) NULL
	error	NVARCHAR(400) NULL